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A Leakage-Controlled Diagnostic Study of How Evaluation Protocols Shape the Apparent Benefits of Hard-Sharing Multi-Task Learning in ADMET Prediction

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Abstract

We examine whether the relative proper-score performance of task-balanced hard-sharing multi-task (MTL) models in ADMET prediction depends on the evaluation protocol and the chemical novelty of test molecules. In a leakage-controlled diagnostic on eight binary ADMET endpoints from the Therapeutics Data Commons, we compare MTL with architecture-matched single-task models under global three-way molecule allocation (70%/10%/20% train/calibration/test) under grouped-random and scaffold-grouped protocols, with five split instances and three seeds; the final confirmatory plan was frozen before the corrected three-way rerun. The multi-task model assigns lower log loss on all eight endpoints under both protocols, and the benefit is larger under scaffold-grouped evaluation (protocol contrast $\Gamma = +0.081$, positive in six of eight endpoints). Under the confirmatory design we find no systematic novelty dependence of the benefit under either protocol. A strict per-model temperature calibration—each temperature fitted on the same trained model’s own calibration partition and applied to its own test partition—attenuates both the raw-probability advantage (post-calibration point estimates -0.003 and $+0.003$) and the protocol interaction (calibrated $\Gamma = +0.006$), indicating that the observed gains primarily reflect temperature-correctable differences in raw logit scale. Label-permutation

and pooled-pretraining controls localize the gain to joint optimization of the shared representation rather than to task-specific label transfer. The positive direction of the protocol interaction was reproduced in a matched-scale graph neural network sensitivity analysis ($\Gamma_{\text{GNN}} = +0.026$). These results indicate that reported multi-task advantages are conditional on evaluation protocol and should be accompanied by calibration diagnostics.

Scientific Contribution: We introduce a leakage-controlled framework for quantifying protocol-dependent multi-task versus single-task proper-score contrasts in ADMET prediction, and show that the multi-task log-loss advantage is larger under scaffold-grouped than grouped-random evaluation and is attenuated to near-zero point estimates by strict per-model temperature calibration.

Keywords: multi-task learning, ADMET prediction, evaluation protocol, calibration, benchmarking

1 Introduction

Multi-task learning (MTL) is widely used in ADMET (absorption, distribution, metabolism, excretion, and toxicity) modeling, and several studies report that multi-task models improve over matched single-task (STL) baselines [Jiang et al. \(2025\)](#); [Shahid et al. \(2025\)](#). At the same time, the molecular machine-learning community has accumulated evidence that evaluation protocols materially shape conclusions: out-of-distribution (OOD) split protocols such as scaffold or cluster splits change absolute performance and its rankings [Fooladi et al. \(2025\)](#); [Guo et al. \(2025\)](#), and recent benchmarks in neighboring fields have shown that strong model claims can depend on evaluation details [Ahlmann-Eltze et al. \(2025\)](#); [Wei et al. \(2026\)](#).

Existing split-protocol studies primarily ask how protocol changes a model’s *absolute* performance. We instead ask how protocol changes the *paired MTL–STL contrast*. The second question does not follow from the first: a harder split could lower both models equally, widen the multi-task advantage, or reverse it. To our knowledge, prior work has not jointly characterized variation in the paired MTL–STL contrast across split protocol, target-task sample size, and continuous target-relative novelty while enforcing global molecule partitions.

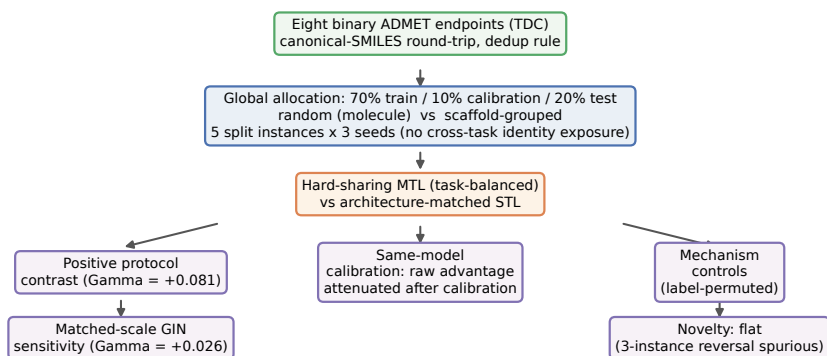


Fig. 1 Experimental framework. Eight binary ADMET endpoints are globally partitioned into train/calibration/test (no cross-task identity exposure) under grouped random and scaffold allocation; hard-sharing MTL is compared against architecture-matched STL via a molecule-level log-loss contrast. The analysis covers the protocol interaction Γ , strict same-model temperature calibration, mechanistic controls (label-permuted MTL, pooled pretraining), a matched-scale GIN sensitivity analysis, and the confirmatory novelty null result.

In this paper we report a leakage-controlled diagnostic whose final confirmatory plan was frozen before the corrected three-way rerun of task-balanced hard-sharing MTL on eight binary ADMET endpoints from the Therapeutics Data Commons [Huang et al. \(2021\)](#). The contributions are the estimand and design, not anticipated findings: (i) a molecule-level log-loss contrast with an explicit endpoint-macro estimand and protocol-contrast estimand; (ii) global molecule allocation controlling cross-task identity exposure, under two valid protocols (grouped random and scaffold-grouped); (iii) a prespecified endpoint-scale manipulation (downsampling); and (iv) a continuous target-relative novelty axis with a preregistered decomposition of calibration and discrimination.

The paper is organized as follows. Section 2 positions the work against prior studies of ADMET MTL, split protocols, and benchmark reliability. Section 3 defines the estimand hierarchy and the leakage-controlled design. Section 4 reports results along the three axes. Section 5 discusses implications and limitations.

2 Related Work

Multi-task ADMET modeling.

Multi-task architectures are common in ADMET prediction. MolP-PC [Jiang et al. \(2025\)](#) trains a multi-view, multi-task framework over 54 endpoints and reports that multi-task learning outperforms single-task training on 41 of 54 tasks; MTAN-ADMET [Shahid et al. \(2025\)](#) proposes an adaptive multi-task network over 24 endpoints with task-specific learning dynamics; industrial-scale studies report benefits of multi-task models on time-split data [Walter et al. \(2024\)](#). Reported gains vary across datasets and settings, and task balancing and endpoint size are themselves known to affect transfer in general MTL [Kearnes et al. \(2016\)](#). These studies report average or per-task performance under a single evaluation protocol and do not condition the reported contrast on protocol choice, endpoint scale, or novelty.

Split protocols and out-of-distribution evaluation.

A growing line of work examines how evaluation protocols change conclusions in molecular property prediction. [Fooladi et al. \(2025\)](#) compare 14 model classes across ten split protocols on eight datasets and find that scaffold splits are not uniformly harder than random splits while cluster-based splits are; [Guo et al. \(2025\)](#) reach similar conclusions on cancer cell lines and show that rank correlations between in-distribution and out-of-distribution performance collapse under cluster splits. [Zheng et al. \(2026\)](#) show that structural-frontier splits expose failures hidden by scaffold splits, and [Guo and Ding \(2026\)](#) find that model-size conclusions depend on protocol difficulty. These studies evaluate single-task models; the paired MTL–STL contrast is not their dependent variable, and none manipulates endpoint size or measures continuous target-relative novelty.

Multi-task transfer under realistic splits.

Studies at the intersection of MTL and split protocols are sparse and reach differing conclusions. [Kearnes et al. \(2016\)](#) observed that multi-task gains were modest and dataset-dependent under temporal splits in industrial ADMET data; [Wu et al. \(2026\)](#) found negative transfer in a two-task setting under scaffold splits, mitigated by safe scheduling; [Walter et al. \(2024\)](#) and CheMLT-F [Mun and Fazli \(2026\)](#) report multi-task benefits persisting under realistic splits. These results are not directly comparable because endpoint sets, architectures, metrics, and molecule-holdout schemes differ. A compact comparison table of these studies on six axes—architecture-matched STL baseline, split protocols, global molecule holdout, endpoint-size manipulation, continuous novelty, and probabilistic calibration—appears in the supplementary material; the table makes the gap concrete: no prior study jointly varies protocol, endpoint scale, and novelty while globally partitioning molecules and reporting calibration-level metrics.

Benchmark reliability and evaluation variability.

The molecular machine-learning community has been repeatedly reminded that evaluation choices can dominate model differences. In single-cell genomics, two major benchmarks reached contradictory conclusions about foundation-model perturbation prediction [Ahlmann-Eltze et al. \(2025\)](#); [Wei et al. \(2026\)](#), with the discrepancy traced to evaluation protocol details. Within molecular property prediction, rank correlations between in-distribution and out-of-distribution performance can collapse under cluster splits [Guo et al. \(2025\)](#), which implies that conclusions drawn from a single benchmark realization—including a single split instance and seed—should be treated cautiously. Our study follows this diagnostic tradition: we measure the conditions under which a design choice (task-balanced hard-sharing MTL) delivers or fails to deliver its reported advantage, with multiple split instances and seeds so that evaluation variability is quantified rather than assumed away.

3 Methods

3.1 Estimand hierarchy

For endpoint e , protocol p , split instance k , training seed r , and test set T_{epk} , let $\hat{p}_{iepk}^{\text{STL}}$ and $\hat{p}_{iepk}^{\text{MTL}}$ denote the predicted probabilities of the single-task and multi-task models for molecule i . We define the molecule-level log-loss contrast

$$d_{iepk} = \ell(y_i, \hat{p}_{iepk}^{\text{STL}}) - \ell(y_i, \hat{p}_{iepk}^{\text{MTL}}), \quad (1)$$

where $\ell(y, p) = -y \log p - (1 - y) \log(1 - p)$; positive d favors the multi-task model.

Endpoint-level and protocol-level aggregates are

$$\Delta_{epkr} = \frac{1}{|T_{epk}|} \sum_{i \in T_{epk}} d_{iepk}, \quad \Delta_p = \frac{1}{8} \sum_e \mathbb{E}_{k,r}[\Delta_{epkr}], \quad \Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}}. \quad (2)$$

Δ_p is the primary endpoint-macro estimand (endpoints weighted equally), and Γ is the protocol-contrast estimand. We treat the eight endpoints as a fixed target set; we do not claim a superpopulation of all ADMET endpoints. Log loss is a strictly proper measure of probabilistic accuracy; ranking discrimination (AUROC/AUPRC) is assessed separately.

3.2 Models and training

The multi-task model comprises a shared 1024–256–128 encoder (ReLU activations, dropout 0.2 after the first hidden layer) and eight independent affine heads mapping the 128-dimensional representation to one logit per endpoint. Model inputs are fixed-length binary fingerprints computed with RDKit’s Morgan algorithm (`GetMorganFingerprintAsBitVect` [RDKit contributors \(2024\)](#)): radius 2, 1024 bits,

default settings (`useChirality=False`, `useFeatures=False`), i.e., the standard non-chiral variant commonly denoted ECFP4; fingerprints were computed once per endpoint from the canonical SMILES and reused across all split instances and seeds. Each single-task model is an independently initialized encoder of the same architecture with one corresponding affine head; we therefore compare against *architecture-matched* single-task models, noting that capacity is matched only on the target-task pathway, not globally.

Training uses *equal task sampling with equal loss coefficients*. Each epoch consists of 64 MTL optimizer updates: eight batches of size $B = 512$ drawn with replacement from each of the eight tasks, each update using that task’s unweighted binary cross-entropy; uniform scheduling gives equal task weighting in expectation. With 60 epochs, each task’s head receives $U = 480$ updates, and the shared trunk receives 3,840 updates in total. Each single-task model receives $U = 480$ updates with $B = 512$ examples under the same sampler; collectively the eight STL models also receive 3,840 updates. Target-task exposure and optimizer-step count are therefore matched between MTL and STL; total examples processed differ (the MTL model additionally consumes auxiliary-task data). Optimization uses Adam ($\eta = 10^{-3}$, weight decay 10^{-4}) with the learning-rate schedule indexed by optimizer step. We scope all conclusions to this task-balanced hard-sharing architecture. As a sensitivity analysis, the protocol contrast was re-estimated with a graph neural network: a three-layer graph isomorphism network (GIN) with hidden width 64, sum pooling (`global_add_pool`), and per-endpoint linear heads, trained on molecular graphs with the same task-balanced schedule (Adam, $\eta = 10^{-3}$, batch size 128, 80 epochs, eight updates per task per epoch) under the same leakage-controlled design and both protocols, at the same scale as the MLP (five split instances $\times$ three seeds). All experiments ran under Python 3.10 with PyTorch 2.13.0 (CUDA 13.0 build) on a single NVIDIA RTX 4070 Ti SUPER GPU; RDKit 2023.09.6 was used for canonicalization, fingerprinting, and scaffold computation.

3.3 Mechanistic controls

Two additional training configurations isolate the mechanism of the contrast, both run under the random protocol at three split instances $\times$ three seeds with all other frozen settings unchanged. (i) *Label-permuted MTL*: for each target endpoint separately, the labels of all other seven endpoints are permuted within each auxiliary endpoint (fixed per-(instance, seed, endpoint) permutations), preserving input exposure, supervision count, and parameterization while removing genuine cross-task label information from the shared trunk; the target-task head still receives true labels. (ii) *Pooled pretrain + STL fine-tune*: a hard-sharing MTL run is used as a shared-trunk initialization, after which each endpoint’s head is re-initialized and the full network is fine-tuned on that endpoint alone with the same task-balanced update schedule. These controls compare (i) label transfer against data exposure and (ii) joint-training gains against transferable-representation gains.

3.4 Leakage-controlled evaluation

A test molecule held out from endpoint e may still appear in the training partition of another endpoint, exposing its structure to the multi-task model through auxiliary labels. We control for this *cross-task identity exposure* by global molecule allocation: each molecule is assigned to a single global partition, and every endpoint inherits that partition. Evaluation is therefore inductive with respect to molecule identity across tasks: a test molecule cannot appear in any auxiliary endpoint’s training partition. No cross-task identity exposure remains under this design. Molecule identity is canonicalized by RDKit canonical-SMILES round-trip (`MolFromSmiles/MolToSmiles`) under a fixed RDKit version, which retains defined stereochemistry in the canonical form. Per endpoint, rows that fail canonicalization are dropped, and rows sharing a canonical SMILES are deduplicated keeping the first occurrence; this is the frozen rule for

duplicate and conflicting labels, applied identically in the development and confirmatory pipelines. Across the eight endpoints, 13 molecules carry conflicting labels (3 in hERG, 10 in BBB_Martins; none elsewhere); excluding all of them changes the endpoint-macro contrasts by less than 2×10^{-4} under both protocols and both raw and calibrated variants, so the deduplication rule does not drive any reported result. We evaluate two protocols under this scheme: grouped random allocation and Bemis–Murcko scaffold-grouped allocation [Bemis and Murcko \(1996\)](#) (with a defined policy for acyclic molecules and empty scaffolds). Five split instances and three training seeds are used; split and seed lists, and their hashes, are frozen. Each instance allocates 70% of the unique-molecule pool to training, 10% to calibration, and 20% to test, and this allocation is global: a molecule’s partition is identical across all endpoints. The calibration partition exists solely for per-model temperature fitting (Section 3.7) and is disjoint from both training and test within each instance. The confirmatory dataset comprises 33,238 unique molecule–endpoint pairs from 17,972 unique canonical molecules; across the 5×3 (instance, seed) combinations this yields 148,008 test observations, with each (instance, seed) combination containing 9,814–9,930 unique molecule–endpoint pairs and no within-combination duplicates. Cross-instance overlap arises because different split instances draw different test partitions; the identity key is the same canonical SMILES used for global allocation.

3.5 Continuous target-relative novelty

For each test molecule we define *target-relative novelty* as one minus the maximum Tanimoto similarity (ECFP4 [Morgan \(1965\)](#), fixed RDKit version [RDKit contributors \(2024\)](#)) to any molecule in the target endpoint’s training partition. Distance to auxiliary-task training molecules is deliberately excluded from the primary definition, so the axis measures distance from the endpoint’s own training support; we preregister a secondary quantity for nearest-auxiliary-training similarity and its interaction with

target-relative novelty, since a molecule far from target training but close to auxiliary training is precisely a candidate transfer case.

3.6 Endpoint scale and downsampling

Endpoint scale is the number of unique target-task training molecules in each split. With eight endpoints, an observational scale association is weak and confounded by endpoint identity; we therefore prespecify a downsampling experiment: large endpoints are subsampled to a fixed target-training size with a frozen sampler (nested subsets, class prevalence preserved, fixed validation and test partitions, auxiliary-task data unchanged, task-balanced schedule held constant). The defensible conclusion from this experiment is an effect of *target-label availability* within the examined endpoints under this training schedule, not a general “scale rather than identity” claim.

3.7 Calibration and discrimination decomposition

A Brier decomposition explains the Brier-score contrast, not the log-loss contrast. We therefore report (i) a Brier reliability–resolution decomposition as a descriptive complement, (ii) calibration intercept and slope, and (iii) AUROC/AUPRC contrasts. A finding of stable AUROC together with worse Brier reliability is reported as evidence *consistent with* a calibration difference, not as proof that the NLL penalty is primarily calibration.

Temperatures are fitted per run, protocol, model, and endpoint, and strictly within the same trained model. Global molecule allocation now assigns each molecule to one of three partitions: 70% training, 10% calibration, and 20% test (random protocol: molecule-level; scaffold protocol: scaffold-grouped). For each (split instance, training seed) run, one scalar temperature T is fitted by minimizing the mean log loss (NLL) on that same trained model’s own calibration partition (disjoint from its training and test partitions; bounded scalar optimization, SciPy `minimize_scalar`, method `bounded`,

$T \in [0.2, 5.0]$) and applied to that same model’s own test partition; no temperature is shared across models. The calibrated molecule-level contrast is recomputed from temperature-scaled probabilities.

3.8 Uncertainty and preregistration

The fifteen split-seed combinations are not fifteen independent replicates: three training seeds share a test set, split instances reuse molecules, all endpoints share one MTL model, and a molecule can carry multiple endpoint labels. Confirmatory inference therefore uses paired contrasts with clustering by standardized molecule, distinguishing split variability from training variability; we report split-level and seed-level dispersion separately. The analysis plan—hypotheses, directional tests, split and seed hashes, estimands and weighting, exclusions, downsampling design, novelty model, multiplicity handling, and analysis code version—was frozen in an immutable record (repository commit 5c2cd1e; protocol/final_confirmatory_plan.md) before confirmatory computation on the leakage-controlled design; the development pass (a per-endpoint split subsequently identified as cross-task-exposed) informed the design and is reported separately as a contaminated stress test in the supplementary material, excluded from confirmatory statistics.

4 Results

We report results for eight binary ADMET endpoints from the Therapeutics Data Commons [Huang et al. \(2021\)](#): hERG, AMES, BBB penetration, P-glycoprotein inhibition, three cytochrome P450 inhibition tasks, and oral bioavailability (Table 1). All conclusions below are restricted to this endpoint set and to task-balanced hard-sharing MTL.

Across the leakage-controlled design (global molecule allocation, five split instances, three seeds), the molecule-level log-loss contrast Δ is positive for every

Endpoint	Test predictions	Unique molecules	Positive rate
hERG	1,938	443	0.69
AMES	21,933	4,914	0.55
BBB_Martins	5,907	1,322	0.76
Pgp_Broccatelli	3,747	832	0.53
CYP2C9_Veith	36,201	8,132	0.33
CYP2D6_Veith	39,246	8,829	0.19
CYP3A4_Veith	37,161	8,350	0.41
Bioavailability_Ma	1,875	416	0.77

Table 1 Dataset statistics. "Test predictions" counts molecule-endpoint test observations across 5 split instances $\times$ 3 seeds (within-(instance, seed) sets contain no duplicates; unique molecules across instances).

endpoint under both protocols: multi-task models assign lower log loss than their architecture-matched single-task counterparts in all 8×2 protocol-endpoint configurations.

4.1 Protocol interaction and overall effect

The endpoint-macro mean contrast is $\Delta_{\text{random}} = +0.321$ under grouped random allocation and $\Delta_{\text{scaffold}} = +0.402$ under Bemis-Murcko scaffold-grouped allocation, with a protocol contrast of $\Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}} = +0.081$ (paired $t_7 = 2.65$, $p = 0.033$). Because the eight endpoints and five split instances are fixed in this study, uncertainty in Γ has three distinct components, which we report separately. Conditional on the observed split instances and training runs, molecule-level resampling gives a narrow interval (95% CI [+0.079, +0.083]); this answers only "how much sampling noise among molecules," not variability across instance realizations. A two-level block bootstrap that resamples split instances first (then aggregates endpoints within them) gives 95% CI [+0.048, +0.112] ($P(\Gamma > 0) = 1.00$), covering split-instance variability. An endpoint-resample bootstrap (resampling the eight Γ_e values) gives 95% CI [+0.026, +0.138] ($P(\Gamma > 0) = 0.998$), covering endpoint heterogeneity; the across-endpoint standard deviation of Γ_e is 0.088. All three intervals exclude zero: the positive protocol contrast remained stable under molecule-, split-instance-, and endpoint-level

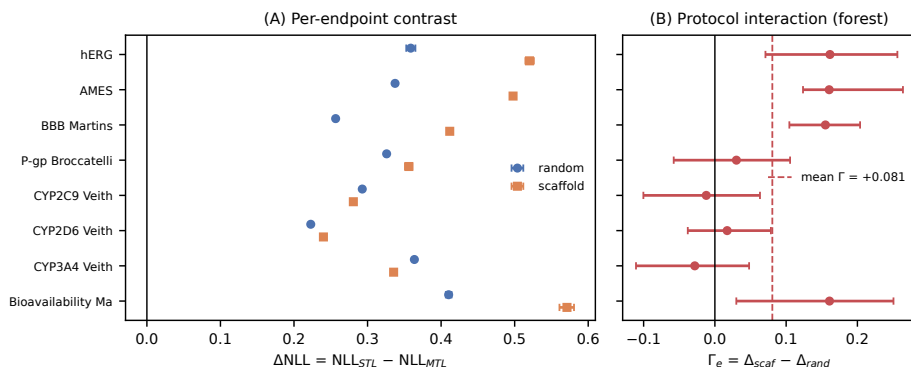


Fig. 2 (A) Per-endpoint log-loss contrast $\Delta = \text{NLL}_{\text{STL}} - \text{NLL}_{\text{MTL}}$ (positive values favor MTL) under grouped random (circles) and scaffold-grouped (squares) allocation, with molecule-level cluster-bootstrap 95% intervals. (B) Per-endpoint protocol contrasts $\Gamma_e = \Delta_{\text{scaf}} - \Delta_{\text{rand}}$ (forest) with endpoint-macro mean (dashed). Six of eight endpoints show positive Γ_e ; leave-one-endpoint-out range +0.069 to +0.096.

resampling within this benchmark panel. The endpoint-resample interval is reported as a sensitivity analysis; the eight endpoints are a fixed target set, so it does not imply an ADMET-endpoint population. The per-endpoint contrasts Γ_e are positive in six of eight endpoints (Figure 2, panel B); the interaction is concentrated among the smaller endpoints in this panel (hERG, AMES, BBB, Bioavailability, each $\Gamma_e \approx +0.16$), while the two largest CYP endpoints show $\Gamma_e \approx 0$ or slightly negative (CYP2C9 -0.012 , CYP3A4 -0.028). A leave-one-endpoint-out analysis leaves the interaction in the range +0.069 to +0.096, so no single endpoint carries the interaction. We present these as concordance and sensitivity analyses for the eight fixed endpoints; the endpoint-dispersion interval does not imply an ADMET endpoint population, and the paired test is a fixed-set diagnostic, not a superpopulation claim. The multi-task benefit is therefore *larger under the scaffold protocol* for the smaller endpoints, and the paired MTL-STL contrast is protocol-dependent: moving from grouped-random to scaffold-grouped evaluation need not shrink the observed MTL advantage. We report the contrast on these eight fixed endpoints; we do not claim an ADMET-wide population effect.

599 4.2 Endpoint-scale moderation and downsampling

600
601 Observationally, the contrast is largest for the smallest endpoints under both proto-
602 cols (e.g., Bioavailability_Ma +0.41; hERG +0.36 under random) and smallest for the
603 largest ones (CYP2D6 +0.22), and the same ordering holds under scaffold allocation
604 (Figure 2). A prespecified downsampling intervention, in which the three CYP end-
605 points were subsampled to 2,000 target-task training molecules with all else held fixed
606 under the current 70/10/20 design, increases the contrast in all three CYP endpoints
607 (CYP2C9 from +0.28 to +0.40; CYP2D6 from +0.24 to +0.38; CYP3A4 from +0.34
608 to +0.53; details in Supporting Information S4). Because the MTL model shares its
609 trunk, the intervention also moves the non-CYP endpoints (mixed direction), so per-
610 endpoint changes are not independent. The consistent CYP increase is consistent with
611 shared representations helping more when target-task labels are scarcer, but we do
612 not claim a causal “scale rather than identity” mechanism from eight endpoints; the
613 observational scale ordering is best read as endpoint-specific, and the downsampling
614 result supports an effect of target-label availability within some endpoints under this
615 training schedule.

626 4.3 Continuous novelty analysis and window test

627
628 Under the confirmatory design (five split instances, both protocols) the contrast shows
629 no systematic dependence on target-relative novelty under either protocol (Figure 3).
630 Under random allocation, the binned contrast moves between +0.296 and +0.304
631 across novelty sextiles with no monotone direction, and within-endpoint novelty slopes
632 are centered on zero with confidence intervals straddling zero for most endpoints;
633 under scaffold allocation the binned contrast moves between +0.309 and +0.366, with
634 the least-novel sextile slightly higher and slopes again straddling zero (four of eight
635 endpoints positive). A hierarchical model of the molecule-level contrast on protocol,
636 novelty, and their interaction (endpoint fixed effects, molecule-clustered bootstrap)
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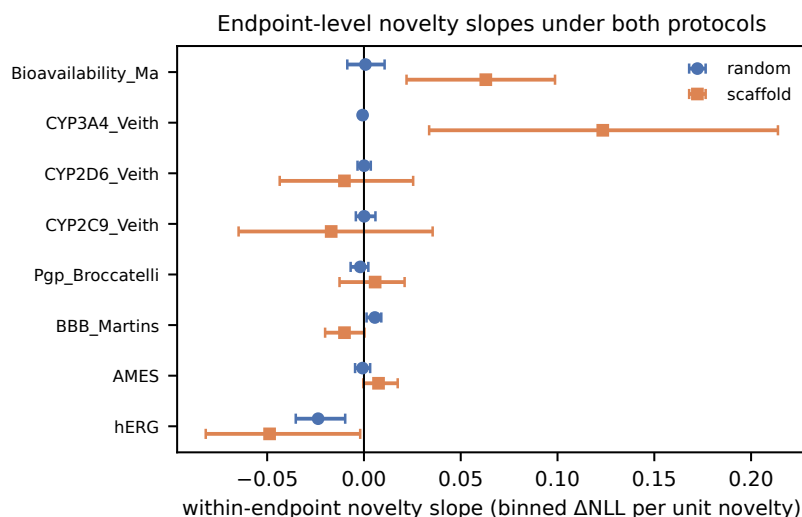


Fig. 3 Novelty slopes under the confirmatory design. Within-endpoint slopes of the binned contrast on target-relative novelty under random (circles) and scaffold (squares) allocation, with unit-level bootstrap intervals. Slopes straddle zero for most endpoints under both protocols; an apparent reversal seen in a three-instance development analysis did not survive five instances (Supporting Information Figure S1).

yields a detectable but negligible interaction ($\hat{\beta} = -0.006$; 95% CI $[-0.010, -0.001]$): a full-unit increase in novelty changes the contrast by only 0.006, less than 2% of the random-protocol macro contrast, so the dependence is not practically meaningful. The preregistered transfer-window hypothesis (an interior peak at intermediate novelty) is therefore not supported under either protocol.

A three-instance development analysis had suggested a striking reversal—declining novelty dependence under random allocation and rising under scaffold allocation—but this pattern did not survive the five-instance confirmatory design (Figure 3; the three-instance curve is shown in Supporting Information Figure S1). We report this as a methodological caution: with only three scaffold-grouped instances, group sampling produced a spurious interaction that the larger design removed, mirroring the lesson from the GNN sensitivity analysis below.

691 4.4 Calibration/discrimination decomposition and robustness

692
693 The contrast decomposes into raw-probability and calibration components. Under the
694 random protocol, Brier scores and reliability components improve for all eight end-
695 points while AUROC is essentially unchanged; under the scaffold protocol, Brier and
696 reliability improve everywhere and AUROC improves in six of eight endpoints (e.g.,
697 hERG $0.76 \rightarrow 0.78$). To test whether the log-loss advantage reflects a durable calibra-
698 tion difference or merely better raw probability estimates, we fitted a per-endpoint
699 temperature T Guo et al. (2017) under the strict per-model design: for each (split
700 instance, training seed) run, protocol, model, and endpoint, one scalar temperature
701 was fitted by minimizing the mean log loss on that same trained model’s own cali-
702 bration partition (disjoint from its training and test partitions) and applied to that
703 same model’s own test partition; no temperature is shared across models. After this
704 calibration, the mean contrast collapses from $+0.321$ to -0.003 under the random
705 protocol and from $+0.402$ to $+0.003$ under the scaffold protocol, and the calibrated
706 protocol contrast is $\Gamma_{\text{cal}} = +0.006$ (per-endpoint calibrated contrasts positive in five
707 of eight endpoints; paired $t_7 = 0.80$, $p = 0.45$). The uncalibrated gains—including the
708 protocol interaction ($\Gamma_{\text{raw}} = +0.081$ vs. $\Gamma_{\text{cal}} = +0.006$)—are therefore attenuated to
709 near-zero point estimates by per-model temperature rescaling (Figure 4), consistent
710 with differences in raw logit scale rather than with ranking information or a calibra-
711 tion advantage that survives recalibration; we report point estimates and note that
712 a formal equivalence analysis against a practical margin is needed before claiming
713 elimination. Class-conditional analysis shows the contrast is carried predominantly by
714 positive-class molecules ($+0.46$ vs. $+0.20$ for negatives under random allocation).

732 4.5 Partition-geometry controls

733
734 Because novelty is defined relative to each endpoint’s own training partition, the
735 scaffold protocol could in principle produce its larger contrast merely by placing
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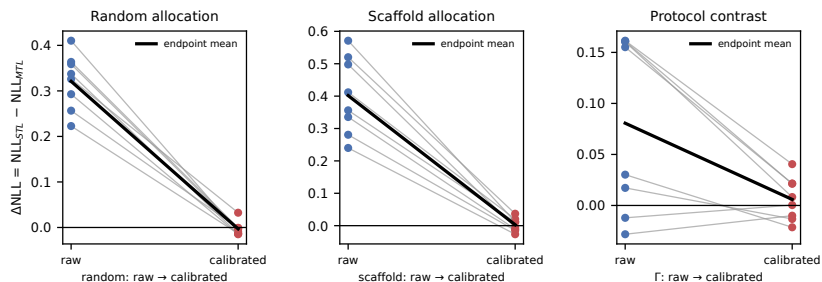


Fig. 4 Temperature-calibration contrast. Mean Δ under both protocols, raw versus strict per-model temperature calibration: the raw advantage (+0.321 random, +0.402 scaffold) attenuates to approximately zero, and the protocol contrast Γ from +0.081 to +0.006 (points: per-endpoint values).

Endpoint	Δ_{rand}	Δ_{scaf}	AUROC _{STL}	AUROC _{MTL}	Γ_e
hERG	+0.36	+0.52	0.836	0.835	+0.16
AMES	+0.34	+0.50	0.876	0.868	+0.16
BBB_Martins	+0.26	+0.41	0.890	0.889	+0.16
Pgp_Broccatelli	+0.33	+0.36	0.922	0.938	+0.03
CYP2C9_Veith	+0.29	+0.28	0.868	0.853	-0.01
CYP2D6_Veith	+0.22	+0.24	0.839	0.829	+0.02
CYP3A4_Veith	+0.36	+0.34	0.877	0.875	-0.03
Bioav_Ma	+0.41	+0.57	0.734	0.733	+0.16

Table 2 Model comparison summary. Δ : log-loss contrast (positive favors MTL); AUROC under random allocation; $\Gamma_e = \Delta_{\text{scaf}} - \Delta_{\text{rand}}$. Full estimates (AUPRC, Brier, reliability, unique molecules) in Table S2.

test molecules at different novelty levels. To test this, we reweighted the random-protocol test molecules so their novelty distribution matches the scaffold protocol’s (percentile-bin reweighting on the pooled novelty grid). The random-protocol contrast is essentially unchanged after matching (+0.3004 vs. +0.3006), and the scaffold advantage ($\Gamma = +0.081$) is not explained by the between-protocol novelty shift. This is consistent with the hierarchical interaction analysis: the protocol-by-novelty interaction is statistically detectable but negligible in magnitude ($\hat{\beta} = -0.006$, 95% CI $[-0.010, -0.001]$), so the larger scaffold contrast is not attributable to a partition-geometry-induced novelty effect.

783 4.6 Decision-level implications

784
785 Proper-score contrasts do not directly translate to practical selection behavior, so we
786 additionally compared MTL and STL on top- k compound prioritization for the three
787 toxicity-relevant endpoints (hERG, AMES, Pgp). Temperature scaling is a monotone
788 transform and therefore does not change ranking, so these comparisons are invariant
789 to the calibration finding. To avoid pooling repeated molecules across split instances
790 and seeds, we computed the top- k statistics separately within each of the 15 (instance,
791 seed) runs and summarized across runs. The per-run differences are small: across
792 the 18 protocol-endpoint- k combinations ($k \in \{50, 100, 200\}$), the mean difference
793 in true toxicants captured in the top k is between -1 and $+2$ molecules, with MTL
794 matching or exceeding STL in the majority of runs for 16 of 18 combinations and
795 no consistent direction of loss in the remaining two (hERG top-50 under random
796 allocation is the single combination where MTL is lower in most runs). Under an
797 illustrative asymmetric-cost utility (false-positive cost 1, missed-toxicant cost c), MTL
798 utility is non-negative relative to STL in 16 of 18 combinations for each of $c \in \{2, 5, 10\}$,
799 i.e., the conclusion does not depend on the cost ratio. The raw log-loss advantage is
800 therefore accompanied by at most small and inconsistent ranking differences, and we
801 do not claim a general prioritization advantage; the decision-level analysis is endpoint-
802 local and descriptive.
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819 4.7 What drives the contrast: mechanistic controls

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821 The MTL and STL systems differ in representation sharing but also in data expo-
822 sure: the MTL model processes auxiliary-task molecules the STL model never sees.
823 To separate label transfer from data exposure, we ran two controls under the random
824 protocol at reduced scale (three split instances, three seeds; all other frozen settings
825 unchanged). First, a label-permuted MTL model: for each target endpoint separately,
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the labels of all *other* seven endpoints are permuted within each endpoint (fixed per- (instance, seed, endpoint) permutations; identical input exposure, supervision count, and parameterization), so the shared trunk cannot extract genuine cross-task label information while the target head still receives true labels. This fully permuted control leaves the contrast essentially unchanged (+0.293 vs. +0.300 for standard MTL), indicating that the raw gain does not require genuine auxiliary-label information under this control. (An earlier control that permuted only a fixed subset of auxiliary endpoints produced unstable negative transfer; we attribute that instability to optimization sensitivity of the permuted configuration rather than to a substantive mechanism, and report only the fully permuted design here.) Second, a shared trunk pretrained jointly on all endpoints and then fine-tuned per endpoint (head re-initialized, full-network fine-tuning) retains only a small fraction of the contrast (+0.045, versus +0.300 for hard-sharing joint training), showing that most of the gain requires the joint training objective itself rather than a transferable shared representation alone (Figure 5). Together the controls localize the effect to joint optimization of the shared representation, with no detectable dependence on task-specific label transfer in this architecture. These controls are endpoint-local and were run at three instances; we report them as mechanistic diagnostics of the main finding.

4.8 Cross-architecture robustness

To test whether the protocol interaction is specific to fingerprint MLPs, we repeated the core comparison with a three-layer graph isomorphism network (GIN) on molecular graphs, under the same leakage-controlled design at matched scale (five split instances, three seeds, architecture-matched STL). The positive direction of the protocol contrast is reproduced: $\Delta_{\text{random}} \approx 0$ (−0.001) and $\Delta_{\text{scaffold}} = +0.025$, giving $\Gamma_{\text{GNN}} = +0.026$ (unit-level bootstrap 95% CI [+0.007, +0.046]; the interval excludes zero), although its magnitude and endpoint pattern differ from the fingerprint MLP (positive Γ_e in

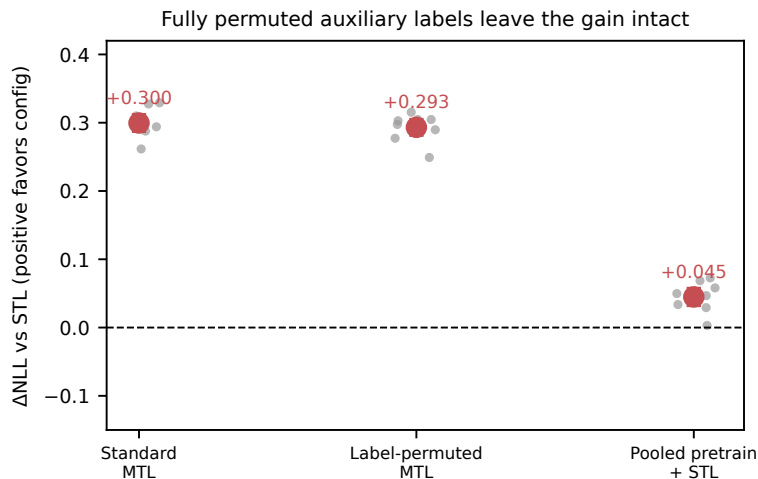


Fig. 5 Mechanistic controls (random protocol, three instances $\times$ three seeds; points: split-seed estimates, error bars: 95% intervals across the nine split-seed values). The standard MTL configuration improves log loss over STL by +0.300; fully permuting all auxiliary labels (target-specific) leaves the contrast essentially unchanged (+0.293), while pooled pretraining followed by per-endpoint fine-tuning retains only +0.045. The controls localize the gain to joint optimization of the shared representation rather than to task-specific label transfer.

four of eight endpoints, driven by hERG and Bioavailability). We read the GNN result as supporting the direction of the interaction in a second representation and model family rather than establishing representation independence. An earlier GNN run with a smaller training budget showed apparent negative transfer; this was an underfitting artifact that disappeared once the training budget matched the MLP configuration.

Endpoint-effect ordering across split instances is not stable (mean pairwise Kendall $\tau = 0.17$, range $[-0.29, 0.57]$), indicating that conclusions about which endpoints benefit most should not be drawn from a single split realization. Full per-endpoint estimates (contrast, AUROC, AUPRC, Brier, reliability, unique molecule counts) appear in Table S2 of the Supporting Information. Table 2 summarizes the per-endpoint model comparison.

5 Discussion

5.1 What the results show

The results qualify, rather than overturn, the commonly reported multi-task advantage in ADMET modeling [Jiang et al. \(2025\)](#); [Shahid et al. \(2025\)](#): under leakage-controlled evaluation, task-balanced hard-sharing MTL provides better raw probability estimates on all eight endpoints and both protocols. The headline finding is the protocol interaction: the paired MTL–STL proper-score contrast is +0.321 under random allocation and +0.402 under scaffold allocation ($\Gamma = +0.081$; paired $t_7 = 2.65$, $p = 0.033$), and the positive contrast was robust to molecule-, split-instance-, and endpoint-level resampling (split-instance block bootstrap 95% CI [+0.048, +0.112]); leave-one-endpoint-out range +0.069 to +0.096, positive in six of eight endpoints and concentrated among the smaller endpoints. Under the confirmatory five-instance design we found no systematic novelty dependence of the benefit under either protocol. The raw-probability advantage—and the protocol interaction itself—does not survive strict per-model temperature calibration, in which each temperature is fitted on the same trained model’s own calibration partition and applied to its own test partition (calibrated $\Gamma = +0.006$, paired $p = 0.45$), so the gains reflect temperature-correctable differences in raw logit scale rather than ranking information or a durable calibration advantage.

5.2 Relation to prior split-protocol and MTL findings

The protocol-dependence finding extends single-task protocol studies [Fooladi et al. \(2025\)](#); [Guo et al. \(2025\)](#); [Zheng et al. \(2026\)](#) to the multi-task contrast, and it helps resolve the apparent tension between studies reporting multi-task benefits under realistic splits [Walter et al. \(2024\)](#); [Mun and Fazli \(2026\)](#) and studies reporting negative transfer [Wu et al. \(2026\)](#); [Kearnes et al. \(2016\)](#): the direction and magnitude of the benefit depend on protocol and on the metric (proper score versus ranking). The

temperature-calibration result adds a methodological caution: raw-probability advantages that disappear under recalibration should be reported as such, and calibration diagnostics should accompany proper-score comparisons. The decision-level analysis revealed only small and heterogeneous ranking differences: MTL matched or exceeded STL in most run-level top- k comparisons, but two protocol-endpoint- k combinations favored STL more often, so we do not infer a general compound-prioritization advantage.

5.3 Methodological lessons

Two results in this study were design-sensitive in ways that reward explicit reporting. First, a three-instance scaffold analysis showed an apparent reversal of the novelty dependence that disappeared under the five-instance design; the GNN sensitivity analysis similarly showed apparent negative transfer before the training budget matched the MLP configuration. Both episodes illustrate that grouped-allocation studies should verify instance and budget adequacy before interpreting pattern-level claims, and they motivate the frozen protocol and the separation of development from confirmatory analyses adopted here. Second, the strict per-model calibration design—fitting each temperature on the same model’s own calibration partition—yields the same qualitative conclusion as an earlier cross-instance calibration shortcut, but the strict design removes a legitimate reviewer objection to the shortcut; we recommend the strict design whenever calibration is part of the evidence.

5.4 Limitations

The endpoint set is limited to eight binary TDC endpoints and one task-balanced hard-sharing fingerprint MLP architecture; other architectures, weighting schemes, and endpoint families may behave differently. The downsampling manipulation increased the contrast consistently across the three CYP endpoints, but because the multi-task model shares its trunk, the intervention also altered the auxiliary endpoints;

the result therefore supports target-label availability as a moderator within this training configuration but does not isolate a general causal sample-size mechanism. Endpoint-effect ordering is unstable across split instances, and with eight endpoints the protocol interaction, while supported by the paired test and bootstrap intervals at all three resampling levels, rests on a small number of endpoint-level observations; we do not claim an ADMET-wide population effect. The temperature-calibration analysis is a point-estimate attenuation analysis; a formal equivalence analysis against a practical margin would be needed to claim elimination, and the calibrated point estimates should be read as approximately zero rather than as proof of exact zero. The top- k decision analysis is endpoint-local and descriptive. We treat all findings as benchmark-local.

6 Conclusion

We presented a leakage-controlled diagnostic with a frozen confirmatory plan of task-balanced hard-sharing multi-task ADMET models. The multi-task proper-score contrast is positive on all eight endpoints under both protocols, larger under scaffold-grouped evaluation ($\Gamma = +0.081$; paired $t_7 = 2.65$, $p = 0.033$; robust to molecule-, split-instance-, and endpoint-level resampling; leave-one-out range $+0.069$ to $+0.096$; positive in six of eight endpoints), and attributable to better raw probability estimates that do not survive strict per-model temperature calibration (calibrated $\Gamma = +0.006$, $p = 0.45$). Under the confirmatory five-instance design we found no systematic novelty dependence of the benefit under either protocol; an apparent reversal in a three-instance development analysis did not survive the larger design.

The positive direction of the protocol interaction was reproduced in a matched-scale GNN sensitivity analysis ($\Gamma_{\text{GNN}} = +0.026$, bootstrap 95% CI $[+0.007, +0.046]$). These results suggest that multi-task proper-score advantages in ADMET modeling

1059 should be reported conditionally on evaluation protocol and accompanied by calibra-
1060 tion diagnostics, and that instance count in grouped-allocation studies should be large
1061 enough to stabilize group sampling. Natural next steps are to extend the diagnostic
1062 to larger endpoint panels, alternative multi-task formulations, and additional OOD
1063 protocols such as cluster or temporal splits.

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1069 **Declarations**

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1072 **Availability of data and materials**

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1074 The dataset(s) supporting the conclusions of this article are available in the
1075 Therapeutics Data Commons repository (<https://tdcommons.ai/>), endpoint identi-
1076 fiers: hERG, AMES, BBB_Martins, Pgp_Broccatelli, CYP2C9_Veith, CYP2D6_Veith,
1077 CYP3A4_Veith, Bioavailability_Ma. The analysis code, global molecule-partition split
1078 manifests, prediction tables, temperature-calibration module, and the full analy-
1079 sis protocol are available in the GitHub repository [https://github.com/Functionhx/](https://github.com/Functionhx/ADMET-MTL-Diagnostic)
1080 [ADMET-MTL-Diagnostic](https://github.com/Functionhx/ADMET-MTL-Diagnostic) (MIT license; frozen submission snapshot: release v1.0.0).
1081

1082 A persistent Zenodo DOI will be assigned from this release at submission.

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1088 **Competing interests**

1089

1090 The authors declare that they have no competing interests.

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1100 **Authors' contributions**

1101

1102 Y.F. conceived the study, designed the experiments, performed the analysis, and wrote
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List of abbreviations	1119
	1120
• ADMET: Absorption, distribution, metabolism, excretion, and toxicity	1121
• AUPRC: Area under the precision-recall curve	1122
• AUROC: Area under the receiver operating characteristic curve	1123
• ECFP4: Extended connectivity fingerprint, radius 4 equivalent (Morgan radius 2)	1124
• GIN: Graph isomorphism network	1125
• MTL: Multi-task learning	1126
• NLL: Negative log likelihood	1127
• OOD: Out of distribution	1128
• STL: Single-task learning	1129
• TDC: Therapeutics Data Commons	1130
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	1135
	1136
	1137
	1138
	1139
References	1140
Jiang W, Zhang L, et al. MolP-PC: a multi-view fusion and multi-task learning framework for drug ADMET property prediction. Chinese Journal of Natural Medicines. 2025; https://doi.org/10.1016/S1875-5364(25)60945-9 .	1141
	1142
	1143
	1144
	1145
	1146
	1147
Shahid S, Maity D, Chakrabarty S. MTAN-ADMET: A Multi-Task Adaptive Neural Network for Efficient and Accurate Prediction of ADMET Properties. ChemRxiv.	1148
	1149
	1150

1151 2025;<https://doi.org/10.26434/chemrxiv-2025-zhrsk>.
 1152
 1153 Fooladi M, et al. Evaluating Machine Learning Models for Molecular Property Predic-
 1154 tion: Performance and Robustness on Out-of-Distribution Data. Journal of Chemical
 1155 Information and Modeling. 2025;65(20):9871–9891. <https://doi.org/10.1021/acs.jcim.5c00475>.
 1156
 1157
 1158
 1159
 1160 Guo J, et al. UMAP-based clustering split for rigorous evaluation of AI models for
 1161 virtual screening on cancer cell lines. Journal of Cheminformatics. 2025;17:94. <https://doi.org/10.1186/s13321-025-01039-8>.
 1162
 1163
 1164
 1165
 1166 Ahlmann-Eltze C, Huber W, Anders S. Deep-learning-based gene perturbation effect
 1167 prediction does not yet outperform simple linear baselines. Nature Methods.
 1168 2025;22:1657–1661. <https://doi.org/10.1038/s41592-025-02772-6>.
 1169
 1170
 1171
 1172 Wei Z, Wang Y, Gao Y, et al. Benchmarking algorithms for generalizable single-
 1173 cell perturbation response prediction. Nature Methods. 2026;23:451–464. <https://doi.org/10.1038/s41592-025-02980-0>.
 1174
 1175
 1176
 1177
 1178 Huang K, et al. Therapeutics Data Commons: Machine Learning Datasets and Tasks
 1179 for Drug Discovery and Development. Nucleic Acids Research. 2021;49:D1153–
 1180 D1160. <https://doi.org/10.1093/nar/gkaa977>.
 1181
 1182
 1183
 1184 Walter M, Borghardt JM, Humbeck L, Skalic M. Multi-Task ADME/PK Prediction at
 1185 Industrial Scale: Leveraging Large and Diverse Experimental Datasets. Molecular
 1186 Informatics. 2024;43(10):e202400079. <https://doi.org/10.1002/minf.202400079>.
 1187
 1188
 1189
 1190 Kearnes S, et al. Modeling Industrial ADMET Data with Multitask Networks. arXiv
 1191 preprint arXiv:160608793. 2016;.
 1192
 1193
 1194
 1195
 1196

Zheng J, Guo C, Wang Z, Liu X. Scaffold splits hide structural-frontier failures in ADMET models. arXiv preprint arXiv:260710729. 2026;.	1197 1198 1199 1200
Guo J, Ding S. Do Larger Models Really Win in Drug Discovery? arXiv preprint arXiv:260426498. 2026;.	1201 1202 1203 1204
Wu S, Wang M, Yu L. Safe Multitask Molecular Graph Networks for Vapor Pressure and Odor Threshold Prediction. Chemometrics and Intelligent Laboratory Systems. 2026; https://doi.org/10.1016/j.chemolab.2026.104548 .	1205 1206 1207 1208 1209 1210
Mun V, Fazli S. CheMLT-F: multitask learning in biochemistry through transformer fusion. Journal of Cheminformatics. 2026; https://doi.org/10.1186/s13321-026-01199-1 .	1211 1212 1213 1214 1215 1216
RDKit contributors. RDKit: Open-source Cheminformatics Software. https://www.rdkit.org . 2024;Version 2023.09.6, accessed 2026-08-03.	1217 1218 1219 1220
Bemis GW, Murcko MA. The properties of known drugs. 1. Molecular frameworks. Journal of Medicinal Chemistry. 1996;39(15):2887–2893. https://doi.org/10.1021/jm9602928 .	1221 1222 1223 1224 1225 1226
Morgan HL. A Generation of Structural Fingerprints for Automated Classification and Search of Chemical Structures. Journal of Chemical Documentation. 1965;5(2):107–113. https://doi.org/10.1021/c160017a018 .	1227 1228 1229 1230 1231 1232
Guo C, Pleiss G, Sun Y, Weinberger KQ. On Calibration of Modern Neural Networks. Proceedings of the 34th International Conference on Machine Learning (ICML). 2017;p. 1321–1330.	1233 1234 1235 1236 1237 1238 1239 1240 1241 1242